Sanjana Kalarickal

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Education

University of California, San Diego
B.S. Computer Science
Graduation: December 2027

Experience

Co-team Lead & Developer , CSES Innovate (Applied Research) Aug 2025 – Present
AIDE: voice-to-code system for students | Python, FastAPI, Electron, TypeScript, Hugging Face

- Cut voice-to-code latency from 3–5 min to 2–5 s on CPU by preloading 3 fine-tuned models; fully local, zero cloud or API cost.
- Built a speech → intent → codegen pipeline routing 5 commands at 98.8% accuracy (500+ held-out utterances) into VS Code.
- Built graceful degradation: code-validity checks on transcripts, auto re-prompting, and escalation to a chat-assistant fallback.
- Demoed live at UCSD CSES Confrence to 200+ attendees: 96% success on 150 unrehearsed prompts. Currently working on system architeture recorvery paper under Prof Colbez.

Software Engineer Intern, Y STEM and Chess Inc. June 2026 – Present
MERN, Agora, Socket.io |

- Built a real-time chess trainer over Socket.io serving randomized puzzles across lessons and competitions, with a context-aware LLM tutor.
- Built live mentor–student sessions in React on Agora (ported from Angular): picture-in-picture mentor overlay, screen share, per-student recordings.
- Refactored an 809-line signup component into 6 tested components, raising coverage from 10% to 87%.

Full-Stack Engineer Intern, Brainworks Academy Mar 2026 – May 2026
React, TypeScript, Express, PostgreSQL/Supabase, Zod, Vitest | Vercel

- Shipped Stripe Payment Element checkout with idempotency keys, processing 55 live orders with zero double-charges.
- Built async Stripe webhook handling with timeouts so slow or duplicate callbacks couldn't corrupt order state.
- Built duplicate-safe user registration with 95% test coverage.

Projects

LitSynth: Autonomous Research Synthesis Agent Jan 2026 – Present
Python, LangGraph, LangChain, SQLite, Docker |

- Built a research agent over **4** paper APIs (arXiv, Semantic Scholar, DBLP, Crossref) extracting citation-grounded claims.
- Parallelized retrieval, extraction, and contradiction checks as LangGraph fan-out, cutting synthesis from **6.5 min to 70 s**.
- Gated output behind a writer/critic loop that retracts claims failing citation-whitelist checks, plus thread-safe cost accounting.

Technical Skills

Languages: Java, Python, TypeScript, JS, SQL, MERN

Frontend: React, Tailwind CSS, Electron,

Backend: REST,FastAPI, Node.js, Express, PostgreSQL,Supabase, MongoDB, SQLite

Cloud: AWS, Vercel, Docker, CI/CD, Git

Testing & ML: pytest, Vitest, PyTorch, Hugging Face Transformers, LangGraph, RAG and vector search (FAISS), Agile/Scrum

Leadership

- Team Lead (CSE 110): Ran an 11-person Agile team through 5 sprints, Designed git workflow and project architecture; Final Approver for 40+ PRs. **Won Best Agile Workflow Award among 50 teams**.

- Founder & President, Girls Who Code(HHS): 50+ members, 100+ workshop and hackathon hours.

- Technical Manager and Tutor at Bored of Boredom : 150+ hours game dev sessions.

Coursework

Database principles,Advanced SWE,SWE with GenAI,Statistical NLP, Recommender system and web mining,Algorithms.